

CrowdVision — Build Plan v9 (Camera-Mesh Final)

Event: Snapdragon Multiverse Hackathon, Qualcomm Bldg N Café, Brookefield, Bengaluru — July 11–12, 2026

Prize target: Top Prize (primary) · Multi-Device Prize (second landing zone, same build)

Supersedes: v8. Headline change: **a 5-feed vision mesh** — 4 personal phones as RTSP camera nodes + 1 scripted surge feed — multiplexed through a single NPU session on the X Elite. Zero new purchased hardware; the doctrine (provided devices + personal phones/laptops only) holds.

Prepared: July 7, 2026.

One-line: An edge-first crowd-safety nervous system: a Snapdragon zone-brain that watches five camera feeds at once on the NPU, sees dangerous density forming minutes early, and physically redirects the crowd in under 2 seconds — no human in the loop, no packet leaving the venue.

Tagline: *See it coming. Stop it early.*

Honest sentence, unchanged: no plan guarantees first place; this plan guarantees no points die for preventable reasons on any of the four criteria.

§0.1 — Changelog [v9]

v9 changes (this revision)

#	Change	Sections touched
1	5-feed vision mesh: 4 personal phones (C1–C4) stream RTSP to the zone-brain; Feed A stays the scripted surge clip (the kill-shot must be deterministic — you cannot stage a real dangerous crowd). Replaces "Surface webcam as Feed 2".	§b, §c, §d, §f, §g, §i, §k, §l
2	Phone role conflict resolved: Phone-H = hotspot and Officer-2 app (compatible roles; hotspot is background). Hotspot	§0.2, §b

#	Change	Sections touched
	backup = any camera phone re-flagged in 60 s (mesh degrades to 3 cams by pre-agreement).	
3	Stream budget specified: 640×480 @ 12 fps, H.264, ~1 Mbps/stream, ~4–5 Mbps aggregate — comfortable on a 5 GHz phone hotspot. Measured in benchmarks.	§d, §k
4	Inference architecture made explicit: one shared QNN session + round-robin freshest-frame scheduler (not 5 parallel sessions, not batching — reasons in §d). New headline Technical-40 claim: sustained 5-feed multiplexed NPU inference.	§d, §k, §Step-3
5	Multi-camera calibration: <code>tools/calibrate.py --camera c1..c4</code> → named profiles in <code>config/cameras.yaml</code> .	§d, §i
6	Gate 3 gets a dedicated gate-lane camera (C4) → real line-crossing counting on that feed; zone-view virtual gate lines remain for uncamera'd gates. Precision correction: C4's intelligence runs on the PC pipeline , not on the UNO Q — the UNO Q still runs no model (no camera in its kit; Constraint 4). Never tell a judge "the Arduino processes a camera."	§c, §f
7	18-hour charging plan: port map + charging cadence in the schedule; camera phones stream only while on charge , screens off.	§g
8	Lane E redefined as Ops & Story with explicit tasks (camera rig + calibration Sat 13:00–15:00, bench runner overnight, fresh-clone tester Sun 06:00, backup Narrator). No demeaning labels anywhere in repo or docs — audited.	§g
9	Phone NPU gap addressed honestly: Gemma 4 E2B sm8750 NPU build cannot be tested July 10 — the OnePlus 15 arrives Saturday 12:00 . Integration is staged pre-event behind a flag; a timeboxed 30-min probe runs Saturday 13:30 . Outcome either way is a benchmark entry + documented decision: FunctionGemma (GPU, specialist) remains the shipped structurer per Constraint 4; E2B-on-NPU is verified capability + the production upgrade path, not a feature swap.	§c, §h, §g, §k, §Step-3

#	Change	Sections touched
10	Demo: Driver visibly scrolls the event log during the kill-shot showing full decision provenance (playbook ID, density, TTT, <code>triggered_by</code>).	§I
11	Multi-device count updated and counted honestly : 9 physical devices, 7 in orchestrated runtime roles + 1 network-infrastructure phone + sim sisters. Inflated counts cost credibility; the honest phrasing is in §c.	§c, §Step-3
12	Sarvam session (Sat 11:30–12:00) assigned a dedicated listener; if an API or on-device Indic model is offered, it upgrades <code>template_fallback.py</code> to trilingual-on-edge — timeboxed adoption rule at G2. Logged as upside, not risk.	§g, §m
13	Demo prop: folded card reading "GATE 3" beside the UNO Q (made from venue paper — doctrine-legal).	§I
14	Bridge RPC de-risking corrected for reality: no UNO Q exists before Saturday, so the pre-event action is App Lab installed on the personal laptop + exact Bridge class/method names pinned from the User Manual/built-in examples + mocked-Bridge test ; the hardware-true Blink + RPC echo is the <i>first</i> Saturday action at 13:00.	§h, §g
15	Compliance: Rules §17(c) social-media silence instruction added; README checklist requires all five members' names+emails; written scope confirmation at orientation now covers both virtual gate lines and the phone camera mesh.	§n, §g, §i
16	RTSP camera app: IP Webcam (Pavel Khlebovich) is freeware, not open-source — factual correction. Compliance is preserved because the app is an <i>input device</i> (like closed-firmware CCTV or Windows itself), not code included in the submission; the pipeline consumes standard RTSP/MJPEG and the README requires only "any RTSP source." An open-source alternative is listed for purists.	§d, §h, §n
17	<code>--sim-all</code> simulates 5 looping file feeds (<code>sim/sim_feeds.py</code>), so judges reproduce the full mesh with zero phones.	§f, §i

#	Change	Sections touched
18	New risk: RTSP instability / phone thermal throttling over sustained streaming → 480p cap, charge-only streaming, per-feed watchdog + honest DEGRADED/LOST states, rehearsal-locked live-feed mix (minimum live config pre-agreed).	§m, §d, §l

v8 changelog (retained for audit trail)

#	Fact (organizer chat)	v8 change
1	Only AI PC, OnePlus, UNO Q provided; bring anything else yourself	Cut router, USB webcam, tripod, USB sticks, printed poster, HDMI, power banks; jobs reassigned to provided/personal devices
2	UNO Q kit has no camera (SKU list)	Gate-node vision cut; virtual gate lines on PC (v9 upgrades Gate 3 to a dedicated phone camera)
3	Modulinos: 20 each for ~50 teams	Distribution sprint at 12:00; zero Modulino dependencies (feature flags)
4	UNO Q is 4 GB variant	MPU headroom noted
5	Criteria re-confirmed	Criteria Balance Sheet added
6	—	QR attendee web view specified as stretch; drones = one sentence

§0.2 — Hardware Doctrine & Role Map [v9]

Provided by the event: Surface Laptop 7 13" (X Elite, NPU v73) · OnePlus 15 (SM8850, Hexagon v81) · Arduino UNO Q 4 GB · Modulino Knob/Buzzer/Thermo *if secured at 12:00* · Cloud AI 100 REST creds · venue tables/power/(A/V TBC).

From us: 5 team members' personal phones, personal laptops, and the cables/chargers that ship with them. Nothing else.

The five personal phones — explicit role assignment (resolves the role conflict):

Phone	Role	Notes
Phone-H	Hotspot LAN + cellular uplink + Officer-2 app instance	Hotspot is a background function; the officer app is lightweight. Both live on one device without conflict. The live uplink-cut beat = toggle Phone-H's mobile data (LAN survives).
Phone-C1	Camera node — Zone B wide shot (the judges)	The live "Zone B is you" beat
Phone-C2	Camera node — Zone C	Live if stable at rehearsal, else looped file (see §I feed-mix lock)
Phone-C3	Camera node — Zone D	Same rule as C2
Phone-C4	Camera node — Gate 3 lane (dedicated gate camera)	Feeds real line-crossing flow counting for G3, on the PC pipeline

Hotspot backup: no sixth phone exists, so the backup is a *pre-agreed degradation*: any camera phone (C3 by default) is re-flagged to hotspot duty in under 60 s using a pre-saved identical SSID/password profile; the mesh drops to 3 cameras and `capture.py` marks the missing feed LOST honestly. The demo survives on the minimum live config (§I).

Camera phone operating rules: RTSP app running with screen off · 640×480 @ 12 fps cap · on charge whenever streaming · positioned by leaning against venue objects (laptops, bottles) — no tripods exist in the doctrine and none are needed at tabletop scale.

Charging port map (18+ h): Surface USB-C #1 → UNO Q (permanent) · Surface USB-C #2 → OnePlus top-ups · Surface USB-A → Phone-H · personal-laptop ports + each phone's own wall charger at venue outlets → C1–C4 continuous. Check-in action: request a table adjacent to mains outlets.

Deployment framing is unchanged and stronger: the judged install assumes provided platforms + "any Wi-Fi hotspot" + "any RTSP source (a phone works)".

§0.3 – Constraint Audit (v6 critiques → resolutions, v9-current) [v9]

#	Constraint	Resolution
1	On-device genuinely justified	Per-model table (§c/§d). Zone-brain vision: 2 s safety budget vs 300–800 ms cloud RTT with jitter on networks that congest when crowds peak; privacy (no frame leaves the venue — now including four phone streams that stay on the local hotspot); battery survival through mains loss.
2	Audio is not a crowd actuator	Visual primary (matrix arrows + RGB); buzzer = steward chirp, named as such; PA integration = deployment path only.
3	Cloud does what the PC cannot	Venue tier: N-zone fusion (1 real cluster + 2 SIM-labeled), trilingual advisories, cross-zone reasoning, post-event report. Never in the safety path — proven live via Phone-H data toggle.
4	Every model beats its simpler alternative	UNO Q still runs no model (no camera in kit; refusing to contrive one). Gemma 4 E2B NPU probe does not displace FunctionGemma for structuring — a 270 M function-calling specialist beats a general 2 B model at emitting one validated call; the probe is capability verification + upgrade path, documented. Predictor stays analytic by choice.
5	Attack response latency	MVP spine unchanged: frame → density → risk → physical gate signal, <2 s, stopwatch-verified, provenance visible in the log live.
6	Phone role field-honest	OnePlus = officer endpoint (GPS, dispatch, FunctionGemma structuring + form fallback). Camera phones = fixed infrastructure sensors (stand-ins for venue CCTV), not officer devices — no VLM anywhere.

Standing honesty rulings: FunctionGemma badged `litert-gpu` (provided artifact is CPU/GPU). If the Saturday probe puts Gemma 4 E2B on the v81 NPU, its benchmark is badged `litert-npu` and the docs state plainly why it is *not* the shipped structurer. The analytic predictor remains the rehearsed "why no fancy model" answer.

§a — The Pitch (one paragraph, the kill-shot)

Every major crowd crush in India — Hathras, the New Delhi railway station, the RCB parade outside Chinnaswamy just one year ago, ten minutes from this building — shares the same anatomy: the density built for over ten minutes, in full view of cameras, and the response took longer than the collapse.

CrowdVision closes that gap. A Snapdragon X Elite zone-brain turns camera feeds into people-per-square-metre on the NPU — five simultaneous streams through one Hexagon session in our demo — forecasts each zone's time-to-danger, and the moment a zone trends toward crush thresholds it acts in under two seconds: flipping an Arduino-powered gate from green to red, lighting LED diversion arrows, and dispatching the nearest officer's OnePlus — with no human in the loop and no video frame leaving the venue. Above it, Qualcomm Cloud AI 100 runs the venue tier: fusing every zone-brain into one picture, drafting PA advisories in English, Hindi, and Kannada, and writing the post-event report — but the safety loop never depends on it, because crowds kill networks at exactly the moment it matters. *The crowd doesn't wait for the network. Neither do we.*

Delivery: first sentence slowly. State it, don't dramatize, move on.

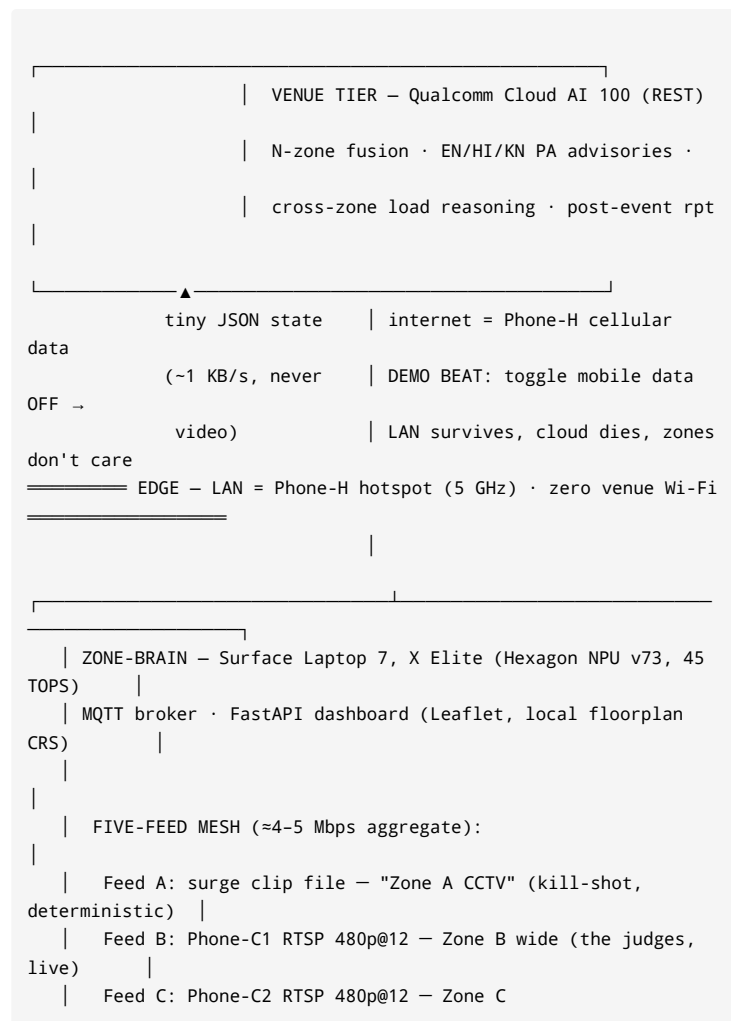
§1 — First-Principles Design (condensed; unchanged in substance)

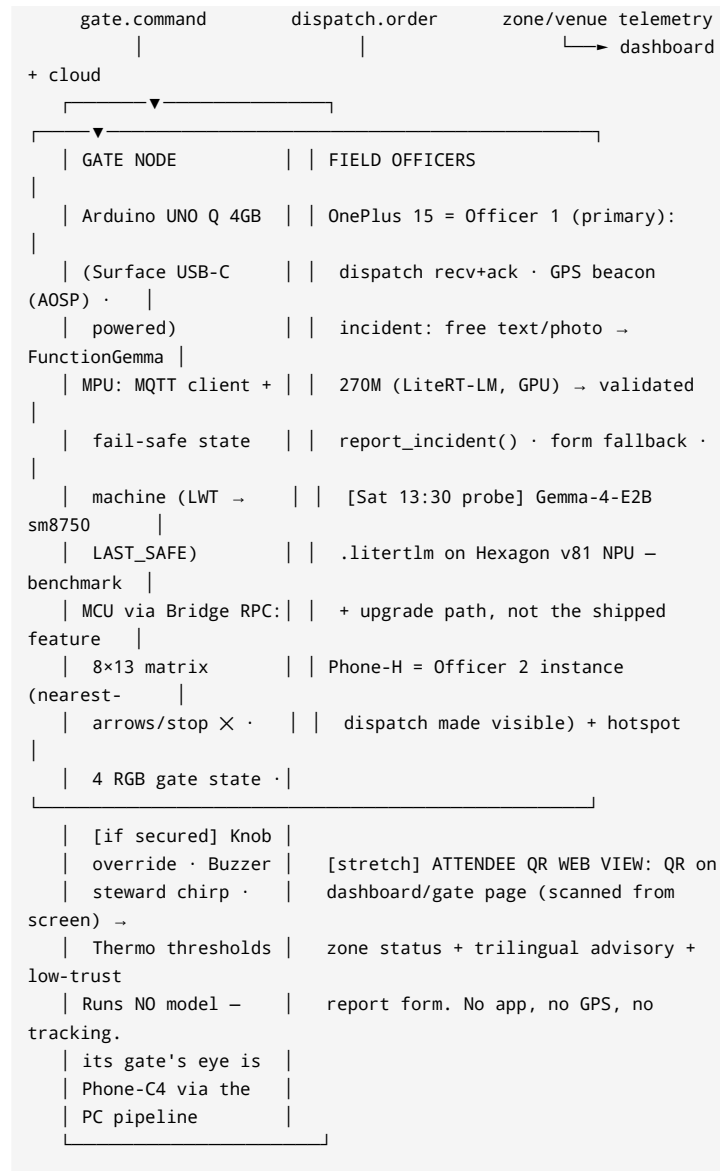
Crowd disasters are a density-and-time problem (Fruin/Helbing): <2 p/m² flows, 3–4 congests, ≥ 5 loses individual control, ≥ 7 collapses. Density accumulates over 5–15 minutes when inflow exceeds outflow. Therefore: (1) detection is a **counting** problem — cameras beat human eyes across many feeds, and the more feeds one brain can watch, the stronger the case; (2) the killer is **response latency** — the intervention that matters is removing the human from the *actuation* loop for pre-approved playbooks, with human override at every level.

```
SENSE (density/m2 per zone, 1 Hz, across a camera mesh)
  → PREDICT (time-to-threshold: EWMA slope + gate flow
conservation)
  → ACT AUTONOMOUSLY (gate signals, diversion arrows, nearest-
officer dispatch)
  → INFORM (operator dashboard w/ override; venue tier:
fusion + trilingual PA)
```

What we deliberately do NOT build: face recognition / identity / re-ID (counts, never people) · any VLM (officer eyes win) · audio sensing or audio-primary actuation · attendee GPS tracking · black-box prediction · internet/cloud/mains dependence in the safety loop · a drone subsystem — the capture layer is source-agnostic RTSP, so an aerial feed is just another URL and gets one sentence.

§b — Architecture [v9]





The kill-shot loop (what the stopwatch measures) [v9 – unchanged path, mesh-aware]

```
sequenceDiagram
    autonumber
    participant F as Feed A (surge clip, Zone A)
    participant PC as Zone-brain (X Elite NPU, 5-feed scheduler)
    participant Q as Gate node (UNO Q)
    participant PH as Officer 1 (OnePlus 15)
    participant C as Cloud AI 100
    F->>PC: frame t (scheduler slot)
    PC->>PC: YOLO INT8 @ shared QNN session (~10-25 ms) → density
4.1/m², slope +0.31/min, TTT 2:40
    PC->>PC: AMBER-RED, playbook P2 (<1 ms) – logged with full
provenance
    PC->>Q: gate.command CLOSE_DIVERT_LEFT (MQTT ~5-20 ms)
    Q->>Q: Bridge RPC → MCU: RGB red, matrix ← arrow, chirp
(~ms)
    Q->>PC: gate.telemetry ACK (actuated_ms)
    PC->>PH: dispatch.order → nearest officer (haversine over
beacons)
    PH->>PC: ack
    PC->>C: zone.state + advisory request (async, OFF the safety
path)
    C->>PC: venue.advisory {en, hi, kn}
    Note over F,Q: Frame → red gate: target < 2 s, ~1.4 s
expected – while four live phone streams keep flowing through the
same NPU session. No human touched anything.
```

§c – Device Role Table [v9]

Device	Role	Why only this device	Honest limits (s: first)
AI PC – Surface Laptop 7, X Elite, NPU v73	Zone-brain. 5-feed density through one shared QNN session + freshest-frame scheduler ; per-camera homography; real gate lines on Feed G + virtual gate lines elsewhere; analytic TTT predictor; playbook autonomy; MQTT broker; dashboard; dispatch. The 2 s safety loop lives here.	Only device with the sustained NPU throughput <i>and</i> RAM/IO to multiplex a camera mesh while hosting broker + dashboard + logs. Cloud fails the 2 s budget	One laptop ≈ one cluster (5 feeds : 480p@12 with headroom). Venu many zone-brain precisely the cloi job.

Device	Role	Why only this device	Honest limits (start first)
		on congested networks and would need four video uplinks instead of 1 KB/s of state. Battery-survives a mains cut.	
4 camera phones (C1–C4, personal)	Fixed camera nodes — stand-ins for venue CCTV: RTSP 640×480@12, screens off, on charge. C1 = Zone B (live judges shot); C2/C3 = Zones C/D; C4 = dedicated Gate 3 lane camera feeding real line-crossing counts.	Only doctrine-legal way to give the demo a <i>real</i> multi-camera mesh; in deployment these are literally replaced by existing CCTV over the same RTSP contract — the pipeline cannot tell the difference, which is the point.	They are sensors; compute: all interleaves runs on the PC. Phones carry honest OK/DEGRADED/ states; a lost fee zone to UNKNOWN hold state, operator alerted) rather than silently guessing
Arduino UNO R4 GB (Surface USB-C powered)	Gate signal + fail-safe endpoint. MCU via Bridge RPC: matrix arrows/stop, RGB state; Knob override, Buzzer steward chirp, Thermo input <i>if secured</i> . MPU: MQTT + LWT-driven LAST_SAFE state machine, chirp-once, auto-rejoin.	Only device physically at the gate at a sub-₹5,000 BOM whose MCU keeps signals deterministic even if Linux hiccups. MPU ↔ MCU Bridge RPC is itself the dual-brain	Runs no model — camera ships in; we refuse to connect (Constraint 4). Its eye is Phone-C4 PC pipeline — not the Arduino processing video. Modulo enhancements, and dependencies.

Device	Role	Why only this device	Honest limits (start first)
		orchestration demo.	
OnePlus 15 – SM8850, Hexagon v81	Officer 1 endpoint. GPS beacon (AOSP LocationManager), dispatch receive/ack + route hint, incident reporting: photo + free text → FunctionGemma 270M → validated <code>report_incident()</code> ; dropdown form = zero-AI fallback. Sat 13:30 timeboxed probe: load <code>gemma-4-E2B-it_qualcomm_sm8750.litertlm</code> on the v81 NPU; if it runs, capture TTFT/tok-s badged <code>litert-npu</code> and document it as the production upgrade path.	Only provided device on an officer's body: GPS, camera, human attached. FunctionGemma automates the dispatcher's structuring step offline; schema validation → misfire is a no-op.	FunctionGemma shipped structured the E2B probe as a 270 M function specialist beats 2 B model at em validated call (C4), and the docs plainly. Probe fails also documented honestly.
Phone-H (personal)	Hotspot LAN + cellular uplink + Officer 2 app instance (makes nearest-dispatch visibly real: two dots, system picks the closer). Also the live uplink-cut beat.	—	Network infrastructure first; Officer 2 is expendable in the anything competent
Qualcomm Cloud AI 100 – REST	Venue tier. Fuses per-zone state vectors from N zone-brains (1 KB/s JSON, never video); trilingual PA advisories (Sarvam swap-in if offered — §g 11:30); cross-zone load answers; post-event replay + report.	Only tier with headroom to serve N zones and heavy multilingual generation without stealing cycles from any zone's saturated NPU. Venue-wide fusion is impossible from inside one zone.	Never in the safe proven live via PI data toggle. <code>template_fall</code> badged <code>fallback</code>
Personal laptops	WSL2 build host, artifact mule (wheelhouse/models/APK), fresh-clone tester, second dashboard	—	Not part of the ju application.

Device	Role	Why only this device	Honest limits (s: first)
	display, charging ports for the mesh.		

Multi-device landing zone [v9 – counted honestly]. Nine physical devices sit on the table: the zone-brain PC, the UNO Q gate node, Officer 1 (OnePlus), four camera nodes (C1–C4), Phone-H, and Cloud AI 100 above them. Of these, **seven play orchestrated runtime roles** (PC, UNO Q, OnePlus, C1–C4, Cloud – plus the Officer-2 instance riding on Phone-H); one is network infrastructure. The choreography is bidirectional on every leg (PC ↔ gate commands/telemetry, phones ↔ PC beacons/dispatch/ack, cameras → PC streams with health states, PC ↔ cloud state/advisory), every leg has LWT-driven failure behavior, the UNO Q adds MPU ↔ MCU Bridge orchestration *inside* one device, and the whole loop is physically observable and stopwatch-timed. Say "seven orchestrated devices across four form factors plus cloud" – judges reward honest counting over inflated counting.

§d – Technology Stack [v9]

Layer	Exact artifact / method	Confirmation / rationale
Camera nodes	Android RTSP/MJPEG streamer app on C1–C4: IP Webcam (Pavel Khlebovich) – freeware, NOT open-source (factual correction to the team note). Compliance stance: the app is an <i>input device</i> (like closed-firmware CCTV, or Windows itself as the OS) – no closed code is included in the submission; the pipeline consumes standard RTSP/MJPEG and the README requires only "any RTSP source." Purist alternative listed in docs: an open-source streamer (e.g., libstreaming-based) if a judge objects – the pipeline doesn't care.	Rules §7c.i concerns pipeline code, not peripherals; stance stated in README Notes
Stream budget	640×480 @ 12 fps, H.264/MJPEG, ~1 Mbps/stream, 4 streams ≈ 4–5 Mbps aggregate on the 5 GHz hotspot – an order of magnitude	§k measures actual throughput + drop rate

Layer	Exact artifact / method	Confirmation / rationale
	under hotspot capacity; leaves MQTT/dashboard untouched	
Capture layer	<code>zone-brain/vision/capture.py</code> : N sources (file/webcam/RTSP) from <code>config/cameras.yaml</code> ; per-feed watchdog — reconnect with exponential backoff, stale-frame detector (>10 s $\Rightarrow$ LOST), OK/DEGRADED/LOST published on <code>cv/camera/{id}/health</code>	v9 risk item 18
Inference architecture	One shared YOLOv8-Det INT8 QNN session (<code>http_performance_mode: burst</code>) + round-robin freshest-frame scheduler (<code>vision/scheduler.py</code>): each feed contributes its newest frame; stale frames are dropped, never queued. Why not 5 parallel sessions: they contend on one NPU and multiply memory. Why not batching: QNN context binaries are fixed-shape and batch>1 kills per-frame latency determinism. One session at ~10–25 ms/frame $\Rightarrow$ ~50–75 inferences/s aggregate $\Rightarrow$ 10–15 effective fps/feed across 5 feeds — the headline Technical-40 claim, measured in $\S k$.	AI Hub guide Track 1; this is the biggest talking point — say it exactly like this
PC vision model	<code>qai_hub_models.models.yolov8_det.export --target-runtime qnn --device "<exact X Elite label>" INT8</code> (+ <code>yolov11_det</code> backup), person class, 60-image real-crowd calibration set	Dev Guide
NPU proof	<code>scripts/verify_npu.py</code> = the guide's <code>get_ep_devices()</code> snippet (never <code>get_available_providers()</code>); output committed to BENCHMARKS.md	AI Hub guide Track 2 step 5
Calibration	<code>tools/calibrate.py --camera c1..c4</code> — click 4 floor points per camera $\rightarrow$ homography + zone polygon per profile in <code>config/cameras.yaml</code> ; <code>--verify</code> overlays the grid live	v9 item 5

Layer	Exact artifact / method	Confirmation / rationale
Gate-flow counting	Hybrid: real gate lines on Feed G (C4's dedicated lane view — higher accuracy, directed in/out counters) + virtual gate lines derived from zone views for uncamera'd gates; both in <code>vision/gatelines.py</code> , method badged in telemetry	v9 item 6
Fall-confirm (should-have)	Fork <code>quic/Pose-Detection-with-HRPOSENet</code> ; aspect-ratio proposes, pose confirms; runs on Feed B slots	Official DevRel sample
PC LLM (should-have)	Genie bundle (open model) via <code>onnxruntime-genai</code> (<code>-genai-qnn</code> does not exist), exported pre-event in WSL2 — offline shift-buddy Q&A	AI Hub guide
Phone models	Shipped: <code>FunctionGemma</code> <code>Mobile_actions_q8_ekv1024.litertlm</code> (LiteRT-LM, GPU backend , badged <code>litert-gpu</code>). Staged behind a flag: <code>gemma-4-E2B-it_qualcomm_sm8750.litertlm</code> + the NPU <code>.so</code> set from the sample app (<code>libLiteRtDispatch_Qualcomm.so</code> , <code>libQnnHtp.so</code> , <code>libQnnHtpV81Skel.so</code> , <code>libQnnHtpV81Stub.so</code> , <code>libQnnSystem.so</code> , <code>libGemmaModelConstraintProvider.so</code>) for the Saturday probe	Dev Guide Track 1 model list + NPU library list
Risk engine	Analytic: EWMA($\alpha=0.3$) @1 Hz → 60 s slope → TTT; flow conservation; hysteresis 10% + 5 s dwell; temp modifier; stale-feed policy: LOST → zone UNKNOWN, gates hold state, operator alerted — never silently guess	Deliberately not ML
Messaging	<code>mosquitto</code> broker on PC (Python <code>amqtt</code> fallback); <code>paho-mqtt</code> (UNO Q); Paho/HiveMQ Kotlin (phones); LWT everywhere	Open source
Dashboard	FastAPI + WebSocket fan-out; Leaflet local-CRS floorplan (zero internet); Chart.js sparklines; per-feed health chips; per-message backend badges	Leaflet/OSM per submission reqs

Layer	Exact artifact / method	Confirmation / rationale
Gate node	App Lab app: <code>python/main.py</code> (MQTT + fail-safe state machine) + <code>sketch/sketch.ino</code> (Bridge RPC; Modulinos behind feature flags)	UNO Q guide
Venue tier	AI Inference Suite REST client + prompts + <code>sim_zones.py</code> (2 SIM zones) + <code>template_fallback.py</code> ; Sarvam upgrade slot : if an API/model is offered at the 11:30 session, the fallback upgrades to trilingual-on-edge — adopt only if integration fits 90 min, decided at G2	Info Guide; §g
Attendee view (stretch)	<code>server/attendee.py</code> : on-screen QR → mobile web page (zone status + trilingual advisory + low-trust report form)	—
Sim harness	<code>python -m crowdvision.sim --all</code> now spawns <code>sim/sim_feeds.py</code> : 5 looping file feeds + virtual gate + virtual officer — judges reproduce the full mesh with zero phones	v9 item 17
License	MIT at root	Rules §8c

Pinned versions; win-arm64 wheelhouse staged on the personal laptop July 8–10; install never touches venue Wi-Fi.

§e — Message Schema [v9 — one new type, two extended fields]

Envelope: { "type": "...", "v": 1, "ts": "2026-07-12T07:41:03.214+05:30", "src": "zonebrain-A", "seq": 4812, "payload": { } }

Rule: every AI-produced message carries `inference_backend`, `latency_ms`, `model_id`; non-AI messages carry provenance (`playbook_id`, `triggered_by`, `provenance`).

Topics: `cv/zone/{id}/density` · `cv/camera/{id}/health` [v9 new] · `cv/gate/{id}/cmd` · `cv/gate/{id}/telemetry` · `cv/officer/{id}/beacon` · `cv/incident/new` · `cv/dispatch/{officer_id}` · `cv/venue/advisory` ·

cv/venue/state · cv/attendee/report (stretch) ·
cv/sys/heartbeat/{device} (retained, LWT).

1. **zone.density.update** (PC → dashboard/cloud, 1 Hz/zone) – [v9] gains
camera_id, transport, fps_effective:

```
{ "zone_id": "B", "camera_id": "c1", "transport": "rtsp",  
  "fps_effective": 11.8,  
  "people_count": 87, "area_m2": 21.0, "density_per_m2": 4.14,  
  "trend_per_min": 0.31, "ttt_red_s": 160, "risk": "AMBER",  
  "flow_check": { "gateline_in_per_min": 42,  
    "gateline_out_per_min": 18,  
    "method": "real-gate-line/c4|virtual-gate-  
line/zone-view", "residual": 0.06 },  
  "temp_c": 33.5, "temp_source": "modulino-thermo|config-  
default",  
  "model_id": "yolov8n-det-int8-qnn",  
  "inference_backend": "qnn-npu-hexagon-v73", "latency_ms": 14.2  
}
```

2. **camera.health** (PC watchdog → dashboard, 0.2 Hz/feed) [v9 new]:

```
{ "camera_id": "c3", "transport": "rtsp", "resolution":  
  "640x480",  
  "fps_effective": 11.6, "drop_rate_pct": 1.8,  
  "last_frame_age_ms": 85,  
  "state": "OK", "reconnects": 0, "note": "OK|DEGRADED|LOST" }
```

Policy: LOST (>10 s stale) ⇒ zone flips UNKNOWN, gates hold state, operator alerted – honesty over silent guessing.

3. **gate.command** (PC → UNO Q; QoS 1, retained, TTL): unchanged from v8 –

```
{ "gate_id": "G3", "action": "CLOSE_DIVERT_LEFT",  
  "allowed":  
  ["OPEN", "CLOSE", "DIVERT_LEFT", "DIVERT_RIGHT", "CLOSE_DIVERT_LEFT",  
  "CLOSE_DIVERT_RIGHT", "SAFE_FLASH"],  
  "reason": "zone B density 4.1 rising 0.31/min, TTT 2:40",  
  "playbook_id": "P2", "triggered_by": "seq:4812", "ttl_s": 120 }
```

4. **gate.telemetry** (UNO Q → PC, 1 Hz; the ACK): unchanged –

```
{ "gate_id": "G3", "state": "CLOSE_DIVERT_LEFT", "actuated_ms":  
6,  
  "bridge_rpc_ms": 4, "override": "NONE", "failsafe_active":
```

```

false,
  "temp_c": 33.5, "modulinos": {"knob": true, "buzzer": true,
  "thermo": true},
  "link_ok": true, "provenance": "deterministic-mcu" }

```

5. **incident.report** (phone → PC): unchanged (FunctionGemma, `litert-gpu`, TTFT reported).
6. **dispatch.order** / 7. **officer.beacon**: unchanged.
7. **venue.advisory** (Cloud → PC/dashboard): unchanged shape; [v9] `inference_backend` may now read `"cloud-ai100"` | `"template-local"` | `"sarvam-edge"` if the Sarvam upgrade lands — badged truthfully either way.
8. **venue.state** / **attendee.report** (stretch): unchanged.

\$f — Feature Tier List [v9] ("L/P?" = reduces Latency or improves Prediction)

MVP — must demo:

Feature	L/P?	Notes
5-feed mesh density on one NPU session (surge clip + C1 live; C2–C4 live per rehearsal lock)	P	Headline: multiplexed sustained NPU inference; per-feed health chips
Hybrid gate-flow counting: real gate lines on C4 (Gate 3) + virtual gate lines elsewhere	P	Method badged per gate
Analytic risk engine: TTT, flow conservation, hysteresis, stale-feed policy	P	Auditable; <code>zones.yaml</code> + <code>cameras.yaml</code>
Autonomous playbooks → physical gate actuation (matrix arrows, RGB)	L	The kill-shot; stopwatch overlay
Gate fail-safe on link loss (LWT → LAST_SAFE, auto-rejoin)	L	Q&A-demoable by dropping UNO Q from hotspot
Officer loop: dispatch/ack, GPS beacon, nearest-officer routing (Officer 2 on Phone-H	L	—

Feature	L/P?	Notes
makes it visible)		
FunctionGemma incident structuring + form fallback	L	Misfire = validated no-op
Dashboard: floorplan, zones/gates/officers/incidents, event log with full decision provenance , feed-health chips, per-gate override	L	It's a URL — any LAN device opens it
Venue tier: trilingual advisory + 1 real + 2 SIM zones + live uplink-cut beat	L	Off the safety path
Sim-all with 5 simulated feeds (<code>sim_feeds.py</code>)	—	Judges reproduce the mesh with zero phones

Should-have (open only if G3 passes): HRPoseNet fall-confirm on Feed B [P/L] · Modulino trio if secured (Knob [L], Thermo [P], Buzzer) · Gemma-4-E2B NPU probe *result write-up + benchmark row* (probe itself is Saturday-13:30 regardless; wiring anything beyond the benchmark is should-have) · officer ETA chips [L] · post-event replay heatmap · Genie shift-buddy.

Stretch: QR attendee web view · Whisper operator voice override · Sarvam-powered trilingual edge fallback (if offered; 90-min timebox) · Cloud what-if UI · OSM zoom-out.

Explicitly cut (README table): face recognition / identity / re-ID · any VLM · audio sensing / audio-primary actuation · attendee GPS auto-linking · chatbot escape guidance · UNO Q vision (no camera in kit — its gate's eye is C4 on the PC pipeline) · E2B replacing FunctionGemma for structuring (Constraint 4: the 270 M specialist wins the job) · drones (one sentence) · WhatsApp/SMS.

§Step-3 — Criteria Balance Sheet (40 / 25 / 20 / 15)

[v9]

Win condition unchanged: **no criterion below ~85% of its weight.**

Technical Implementation — 40 pts (target 36+).

Earns [v9]: **the mesh claim** — "five live streams multiplexed through one Hexagon NPU session, ~50–75 INT8 inferences/s sustained, 10–15 effective fps per feed,

burst profile, verified via `get_ep_devices()` and a 10-minute soak with zero thermal decay" · NPU-vs-CPU side-by-side · <2 s frame → gate loop measured live · per-workload power profiles with battery-delta evidence · [v9] the OnePlus NPU probe result — success gives a `litert-npu` generative benchmark on the phone (TTFT + tok/s on Hexagon v81); failure gives a documented, honest attempt with the exact error — either outcome demonstrates engineering rigor. Typically lost to: silent CPU fallback · one blob number for generative models · burst-everything energy mismatch · unverifiable claims · **[v9] flaky live streams read as instability** — countered by the watchdog's honest DEGRADED/LOST states and the rehearsal-locked feed mix.

Application Use-Case & Innovation — 25 pts (target 23+).

Earns: autonomy attacking response latency · Bengaluru-grounded opening · physical, watchable actuation · privacy-by-design (now including: four phone streams that never leave the local hotspot) · fail-safe UX · trilingual advisories. [v9] The mesh strengthens the deployment story: "in production these four phones are your existing CCTV — same RTSP contract, the pipeline cannot tell the difference." Countermeasures unchanged: the L/P column; the deliberate-cuts list presented, not hidden.

Deployment & Accessibility — 20 pts (target 18+).

Earns: `--sim-all` now reproduces the **full 5-feed mesh** with zero phones · 3-command quickstart · prebuilt APK/models/wheels in Releases · provided-hardware doctrine ("any Wi-Fi hotspot, any RTSP source") · fresh-clone test Sun 06:00. [v9] Watch-item: camera setup must not complicate the judged install — it doesn't, because cameras are optional inputs; the README's judge path never asks for a phone.

Presentation & Documentation — 15 pts (target 14+).

Earns: rehearsed 5-min script with roles + 3:30 cue · auto-embedded benchmarks · MESSAGES/ARCHITECTURE docs · honest badges. [v9] New beat: the Driver scrolls the decision log live — provenance as theatre.

Multi-Device Prize (second landing zone) [v9]. The rubric scores cross-device coordination, seamless UX, orchestration depth, usefulness, reliability/polish. The honest claim: **seven orchestrated devices across four form factors plus cloud** (PC, UNO Q, OnePlus, C1–C4, Cloud AI 100; Officer-2 rides Phone-H; Phone-H itself is infrastructure) — bidirectional on every leg, LWT failure

choreography, intra-device MPU ↔ MCU Bridge orchestration, one stopwatch-timed physical loop. Count honestly; judges punish inflation.

§g — Hour-by-Hour Schedule (24 h + morning actions, pre-agreed gates) [v9]

Lanes (5 members): **A** PC vision+engine · **B** UNO Q gate node · **C** phone apps (officer + camera nodes) · **D** dashboard+MQTT+cloud · **E — Ops & Story**: camera rig + calibration (Sat 13:00–15:00), bench harness runner (overnight), fresh-clone tester (Sun 06:00), docs, the clock, **backup Narrator**. Primary Narrator = D (hands-free during the demo). No demeaning role labels exist in any repo file or doc — audited at G5.

Standing rules: decisions were made July 7 · two 20-minute attempts per bug, then the workaround · nothing new after G4.

Time (IST)	Work	Gate & fail branch
Sat 10:00–11:00	Check-in. [v9] Written scope confirmation with organizers covering BOTH architectural deltas from the Phase-1 proposal: (a) virtual/real gate-line counting, (b) the personal-phone camera mesh (Rules §7.b.ii). Request an outlet-adjacent demo table. Ask: demo A/V + Cloud AI 100 endpoint/model/key.	—
11:30–12:00	[v9] Sarvam session — D attends as dedicated listener. Capture: any API/key offer, any on-device Indic model, license terms. If offered → G2 decides adoption (90-min timebox, upgrades <code>template_fallback</code> to trilingual-on-edge; slot <code>sarvam-edge</code> badge).	—
12:00–13:00	Lunch + device distribution sprint : E signs out UNO Q + one Knob/Buzzer/Thermo immediately (20 each event-wide). Team Lead signs loaner.	—

Time (IST)	Work	Gate & fail branch
13:00–14:00	Phone-H hotspot up (C3 pre-cloned as backup profile). [v9] First action on real hardware: UNO Q Blink + Bridge RPC echo (B) – the pinned API names from Sh meet the board. Surface: wheelhouse install + <code>verify_npu.py</code> . OnePlus: APK install. Cloud: one REST round-trip. E+C: camera rig – IP Webcam configured on C1–C4 (480p/12fps, screen off, on charge), RTSP URLs into <code>cameras.yaml</code> , all four feeds visible in <code>capture.py</code> .	G0 (14:00): all devices on LAN, NPU visible, cloud reachable, ≥ 3 of 4 camera feeds stable, Bridge echo passed. Red → mentors/Discord; lanes continue in sim.
13:30–14:00 (parallel, C)	[v9] Timeboxed 30-min OnePlus NPU probe: load <code>gemma-4-E2B-it-qualcomm-sm8750.litertlm</code> with the NPU <code>.so</code> set behind the app flag. Success → capture TTFT/tok-s, badge <code>litert-npu</code> , done. Failure → screenshot the exact error, done. Either way the probe ends at 14:00 and C returns to the officer app. No wiring into features today.	Result recorded in BENCHMARKS.md either way.
14:00–17:00	A: capture → shared-QNN-session → scheduler on Feed A + C1, per-frame ms measured. E: per-camera calibration (<code>calibrate.py --camera c1..c4, --verify</code> overlay). B: gate state machine skeleton, Modulino flags. C: officer app wiring (MQTT, dispatch, beacon). D: broker + dashboard + feed-health chips.	G1 (17:00): detection on NPU (<40 ms/frame) AND ≥ 3 feeds flowing through the scheduler. Fail model → backup export; fail feeds → demo minimum is Feed A + C1 + C4, others become looped files (pre-agreed). CPU-with-honest-badge deadline 19:00 as before.
17:00–20:00	A: homography per camera + density + tracker + real gate lines on Feed G, virtual elsewhere . B: fail-safe timer +	—

Time (IST)	Work	Gate & fail branch
	knob override. C: FunctionGemma + report screen + form. D: zones/gates/officers/log with provenance fields rendered. E: bench harness dry run; charging audit #1 (all C-phones on charge, temps checked by hand). (Dinner 19:30–20:15, two shifts.)	
20:00–21:00	Integration hour: full loop in sim (5 sim feeds → virtual gate → dashboard).	G2 (21:00): end-to-end passes in sim. Also: Sarvam adopt/skip decision (adopt only if D estimates ≤ 90 min and core is green). Fail → pre-agreed cut: 3 feeds, single zone focus, one gate.
21:00–24:00	A: TTT predictor + flow conservation + hysteresis + stale-feed policy + playbooks. B: real gate end-to-end from live density. C: nearest-dispatch + ack chain; Officer 2 on Phone-H. D: cloud advisory + venue view + uplink-cut handling (+ Sarvam if adopted). E: surge clips final; stopwatch overlay; charging audit #2 (23:00) .	—
24:00	Midnight rehearsal: one full live run, all feeds, rough stopwatch number.	G3 (00:30): loop stable & p95 < 2 s with ≥ 4 of 5 feeds live? YES → should-haves open (fall-confirm → A, Thermo/Buzzer → B, replay → D). NO → converge on the core; should-haves die without discussion.

Time (IST)	Work	Gate & fail branch
00:30–03:00	Should-haves or hardening (reconnects, QoS audit, watchdog tuning, badge polish). E: BENCHMARKS run 1 — incl. the 5-feed sustained soak (venue is quiet; numbers are clean). Charging audit #3 (02:30) .	G4 (03:00): FEATURE FREEZE.
03:00–06:00	Two 90-min sleep pods; watch runs the 2 h soak (5 feeds streaming — this doubles as the RTSP endurance test). Camera phones remain on charge, screens off.	—
06:00–09:00	E: fresh-clone test on the personal laptop, README-only — fix docs, not code. Final benchmarks (300-frame NPU vs CPU, 5-feed aggregate, e2e ×50, FunctionGemma ×20, battery deltas, hotspot throughput + drop rate). Record the 2-min backup video → desktop + a personal phone. Charging audit #4 (08:30) — top up every phone to >80% before demos.	G5 (09:00): README reproducible · benches embedded · APK+models in Releases · MIT · all five names+emails · no demeaning labels anywhere. Red → triage: README > run scripts > benchmarks > polish.
09:00–11:00	Demo rehearsal ×2, timed. [v9] Feed-mix lock at the second rehearsal: minimum live config = Feed A (clip) + C1 (judges) + C4 (gate lane); C2/C3 stay live only if zero dropouts across both rehearsals , else they run as looped files with honest <code>transport: "file"</code> badges.	—
11:00–11:45	Repo freeze: tag <code>v1.0</code> , push, public-access check from a phone in incognito, Releases verified.	—
11:45–12:15	SUBMIT the Microsoft Form; screenshot confirmation. Never later than 12:15.	G6.
12:15–13:00	Reset demo station; phones repositioned + charging; "GATE 3" card folded and	—

Time (IST)	Work	Gate & fail branch
	placed; one silent dry run; lunch.	
13:00–16:00	Demos. Narrator = D (E backs up), Driver = A, B at the gate node, C holds the OnePlus. 16:00 device return per loaner checklist.	—

§h — Pre-Event Critical Path (July 8–10) [v9]

July 8 (exports day — morning start; AI Hub queues on shared hardware):

- AI Hub tokens ×2; record the exact X Elite device label into `docs/DEVICES.md`.
- Kick off `yo1ov8_det` INT8 QNN export (+ `yo1ov11_det` backup) with the **60-image real-crowd calibration set** (assembled first; never random arrays).
- Kick off the Genie bundle export (open model, WSL2, overnight).
- Download FunctionGemma `Mobile_actions_q8_ekv1024.litertlm` and [v9] `gemma-4-E2B-it-qualcomm-sm8750.litertlm` + the NPU `.so` set from the **LiteRT-LM sample app** — staged for the Saturday probe.

July 9 (device-code day):

- Officer app: Kotlin skeleton, LiteRT-LM + FunctionGemma, `assembleRelease` APK. [v9] **E2B probe path built behind a flag** (model path + NPU backend config + TTFT/tok-s logging) so Saturday's probe is a toggle, not an integration.
- Gate node: full App Lab app with Modulino feature flags; Python side tested with mocked Bridge.
- [v9] **Bridge RPC de-risking (highest-probability venue failure point)**: install Arduino App Lab on the personal laptop; open the built-in Bridge examples and the UNO Q User Manual; **pin the exact Bridge class/method names** into `gate-node/` (the guide itself warns helper APIs differ by App Lab version); write the mocked-Bridge test against those names. Note honestly: a hardware-true test is impossible before Saturday — the board arrives at 12:00 — which is why Blink + RPC echo is the first 13:00 action.
- Dashboard + `--sim-all` end-to-end with `sim_feeds.py` (5 looping file feeds) on the personal laptop — the judges' path works today.

- Hotspot rehearsal with the exact phones; `devices.yaml` written; laptop ↔ laptop transfer timed.

July 10 (stage + freeze day):

- **[v9] Camera mesh dress rehearsal at home:** IP Webcam installed on C1–C4 (settings preset: 640×480, 12 fps, screen-off streaming); all four RTSP URLs into `cameras.yaml`; run the real 4-stream mesh + Feed A through the full pipeline on the personal laptop for 60 minutes on the actual Phone-H hotspot; capture drop rates; per-camera calibration profiles created with `calibrate.py` (re-verified at the venue table Saturday).
- Full dry run in sim, fresh clone, README-only.
- Stage on the personal laptop: models (YOLO ×2, FunctionGemma, **E2B + NPU .so set**), Genie bundle, wheelhouse, APK, clips, `git bundle`, floorplan PNG.
- Send organizers the scope-confirmation email **naming both deltas: gate-line counting and the phone camera mesh** (Rules §7.b.ii trail). Re-read loaner terms. Charge everything.
- Rehearse the pitch twice against 5:00.

Hard cannot-do-at-venue list: AI Hub exports · Genie bundle export · win-arm64 wheel fetching · APK signing setup · sourcing calibration images/footage · **[v9] camera-app installation and mesh shakedown** (venue time is for re-verification, not first contact).

§i — Repo Structure [v9]

```
crowdvision/
├── README.md           # §j — judges' 5-minute path first
├── LICENSE             # MIT
├── docs/              # ARCHITECTURE.md · MESSAGES.md ·
BENCHMARKS.md (auto-embedded)
|                       # DEMO.md · DEVICES.md (labels,
IPs, power-profile rationale)
├── zone-brain/
|   ├── vision/        # capture.py (file/webcam/RTSP +
per-feed watchdog)
|   |                 # scheduler.py (shared-session
round-robin, freshest-frame) [v9]
|   |                 # detect_qnn.py · homography.py ·
density.py · tracker.py
|   |                 # gatelines.py (real + virtual
```

```

gate lines)                                [v9]
|   └─ engine/                             # risk.py
(EWMA/TTT/hysteresis/stale-feed policy) · flow.py · playbooks.py
|   └─ server/                             # app.py (FastAPI+WS) ·
mqtt_bridge.py · attendee.py (stretch)
|   |                                     # static/ (dashboard: floorplan,
feed-health chips, provenance log)
|   └─ llm/                               # genie_shiftbuddy.py (should-
have)
|   └─ scripts/                           # setup.ps1 · verify_npu.py ·
download_models.py (--local) · run_demo.ps1
|   └─ bench/                             # detect_bench.py · mesh_bench.py
(5-feed sustained)                        [v9]
|   |                                     # e2e_bench.py · net_bench.py
(hotspot throughput/drops)               [v9]
|   |                                     # power_delta.ps1
└─ gate-node/                             # UNO Q App Lab app (pinned Bridge
API names)                               [v9]
|   └─ app.yaml
|   └─ python/ (main.py, requirements.txt)
|   └─ sketch/ (sketch.ino, sketch.yaml)
└─ field-app/                             # Kotlin: FunctionGemma (GPU,
shipped) + E2B NPU probe flag [v9]
└─ venue-tier/                             # aisuite_client.py · prompts/ ·
sim_zones.py · template_fallback.py
|   |                                     # sarvam_adapter.py (empty slot
until/unless offered)                 [v9]
└─ sim/                                   # sim_feeds.py (5 looping file
feeds)                                [v9]
|   |                                     # sim_gate.py · sim_officer.py ·
replay.py · tests/
└─ config/                               # zones.yaml · cameras.yaml
(c1..c4 homography profiles)           [v9]
|   |                                     # playbooks.yaml · devices.yaml ·
.env.example
└─ tools/                                # calibrate.py (--camera cN, --
verify) · make_calibset.py             [v9]

```

Sj – README Template [v9]

```

# CrowdVision – autonomous crowd-crush prevention on the
Snapdragon edge
*Snapdragon Multiverse Hackathon 2026, Bengaluru · Top Prize

```

submission*

Crowds form dangerous densities minutes before humans react. CrowdVision watches five camera feeds at once on a Snapdragon X Elite NPU – one shared Hexagon session multiplexing every stream – forecasts each zone's time-to-danger, and physically redirects the crowd (Arduino gate signals + nearest-officer dispatch on a OnePlus 15) in under 2 seconds, fully on the edge. Qualcomm Cloud AI 100 adds a venue tier (multi-zone fusion, English/Hindi/Kannada PA advisories, post-event report) that the safety loop never depends on.

Team

Name	Email
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<member 1> (Team Lead)	<email>
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<member 2>	<email>
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<member 3>	<email>
------------	---------

<member 4>	<email>
------------	---------

<member 5>	<email>
------------	---------

(All Eligible Individuals listed – Rules §7c.ii.)

Judges: the 5-minute path (zero hardware, zero phones)

```
git clone https://github.com/<org>/crowdvision && cd
```

```
crowdvision
```

```
python zone-brain/scripts/download_models.py    # or --local
```

```
<path>
```

```
python -m crowdvision.sim --all
```

Open <http://localhost:8000> – five simulated camera feeds, live zones, a virtual

gate flipping states, a virtual officer acknowledging dispatches.

Real hardware

replaces sim components on the same MQTT topics with zero code changes.

Full setup from scratch (provided hackathon hardware + any phones)

0. Network: any Wi-Fi hotspot. IPs in `config/devices.yaml`.

1. AI PC (X Elite, Windows ARM64): `zone-brain/scripts/setup.ps1`, then

```
verify_npu.py – proves the QNN EP via get_ep_devices() (not  
get_available_providers()); plugin EP in ORT 2.x). Run:
```

```

run_demo.ps1.
2. Camera nodes (optional – sim feeds otherwise): any RTSP source
at
    640×480 @ 12 fps. We used Android phones running an RTSP
streamer app;
    the pipeline consumes standard RTSP and does not depend on any
specific
    app. Calibrate each: python tools/calibrate.py --camera c1
(repeat c2..c4);
    profiles land in config/cameras.yaml.
3. Gate node (Arduino UNO Q, powered from the PC's USB-C):
    arduino-app-cli app start ./gate-node # Modulinos optional,
auto-detected
4. Officer phone (Android): adb install Releases/field-app.apk.
FunctionGemma
    fetched on first run or sideloaded. A second Android = second
officer.
5. Venue tier (optional – fully functional without it):
AISUITE_ENDPOINT /
    AISUITE_KEY in .env.

## Where the intelligence runs (edge-majority statement – Rules
$7c.iv)
| Function | Device | Backend |
|---|---|---|
| 5-feed density + gate-line flow | X Elite | Hexagon NPU v73,
INT8, one shared QNN session, burst |
| Risk prediction | X Elite | analytic – deliberate; safety logic
must be auditable |
| Gate actuation + fail-safe | UNO Q | deterministic MCU via
Bridge RPC (no model; no camera ships in the kit – its gate's eye
is a dedicated camera feed processed on the PC) |
| Incident structuring | OnePlus 15 | FunctionGemma 270M, LiteRT-
LM, GPU (provided artifact is CPU/GPU – labeled honestly) |
| Phone NPU capability | OnePlus 15 | Gemma-4-E2B sm8750 build
probed on Hexagon v81 – result + numbers in BENCHMARKS.md;
documented as the production upgrade path |
| Venue fusion + trilingual PA | Cloud AI 100 | REST – additive
tier, never in the safety path |

## Benchmarks
docs/BENCHMARKS.md (auto-generated): NPU-vs-CPU per-frame · 5-
feed sustained
aggregate inferences/s + effective fps/feed · e2e frame→gate
p50/p95 ·
FunctionGemma TTFT + tok/s · hotspot throughput + RTSP drop rate

```

```
.
battery-delta per power profile · committed verify_npu.py output.

## Tests · Notes · References
pytest sim/tests verifies the message loop headless. Notes:
privacy by design –
counts, never identities; no faces stored; camera streams never
leave the local
network; the camera app is an input device, not code included in
this
submission (the pipeline consumes standard RTSP). References:
Fruin LOS bands;
Qualcomm AI Hub docs; LiteRT-LM docs; forked official DevRel
samples.

## License – MIT
```

§k – Benchmark Plan (exact numbers · device · method) [v9]

All numbers captured on the actual demo hardware by `bench/` scripts emitting JSON; `docs/BENCHMARKS.md` embeds the tables automatically – no hand-typed numbers, no cloud compile-time estimates.

#	Metric	Device	Method
1	Detection latency mean/p50/p95/p99, NPU vs CPU side-by-side	X Elite	3 warmup + 300 timed frames @640² INT8; once QNN EP (burst), once CPU EP
2	[v9] 5-feed sustained mesh – aggregate inferences/s + effective fps per feed + per-stage breakdown (capture/schedule/infer/track+lines/decide)	X Elite	<code>mesh_bench.py</code> : 10-min soak, Feed A + 4 RTSP live, one shared QNN session, scheduler counters
3	[v9] Network: hotspot throughput + per-stream RTSP drop rate + reconnect count	Phone-H LAN	<code>net_bench.py</code> : 10-min window during #2;

#	Metric	Device	Method
			watchdog counters
4	End-to-end frame → gate-actuated p50/p95	PC + UNO Q	50 playbook fires; frame-ts → gate ACK minus <code>actuated_ms</code> ; single-clock RTT/2 method, stated
5	Gate actuation internals	UNO Q	<code>bridge_rpc_ms</code> ×100 + <code>actuated_ms</code> per command
6	FunctionGemma TTFT + tokens/sec + e2e parse latency + schema-valid rate	OnePlus 15	20 scripted + 5 free-form; balanced profile; badged <code>litert-gpu</code>
7	[v9] Gemma-4-E2B NPU probe — loads? TTFT + tok/s on Hexagon v81	OnePlus 15	Saturday 13:30 timeboxed probe via the app flag; success → numbers badged <code>litert-npu</code> ; failure → exact error screenshot
8	Energy: battery-delta per power profile	X Elite	<code>powercfg /batteryreport</code> before/after two 10-min runs: burst vs balanced (mesh running)
9	NPU proof artifact	X Elite	<code>verify_npu.py (get_ep_devices())</code> ;

#	Metric	Device	Method
			raw output committed, timestamped
10	Venue-tier RTT distribution	Cloud AI 100	30 advisory calls, wall-clock
11	(if in) HRPoseNet fall-confirm latency · Genie shift-buddy TTFT + tok/s	X Elite	Same harness patterns

Power-profile rationale, verbatim in docs and on the numbers slide: *sustained real-time multi-stream video → burst (pins NPU clocks, kills frame-to-frame variance); single interactive inference → balanced; background sensing → efficiency. Chosen per workload, measured, defensible.*

§I – 5-Minute Demo Script [v9] (kill-shot at 1:15; props: provided hardware + one folded card)

Stage: Surface center; UNO Q beside it with a **folded venue-paper card reading "GATE 3"** [v9] — zero cost, makes the actuation beat legible to non-technical judges at a glance. Phone-C1 leaning against the personal laptop, framing the judges (Zone B). Phone-C4 aimed along the tabletop "gate lane" past the UNO Q. C2/C3 positioned per the rehearsal feed-mix lock (live, or looped files with honest **file** badges). All camera phones on charge, screens off. Dashboard on the demo display if provided; otherwise the Surface screen + the URL offered to any judge's device.

Roles: Narrator = D (E backs up) · Driver = A · B at the gate node · C holds the OnePlus.

Clock	Beat	Script + action
0:00–0:30	Hook	"Hathras. New Delhi station. The RCB parade, ten minutes from here. Same anatomy every time: the density built for over ten minutes on camera, and the response took longer than the collapse. CrowdVision closes that gap."

Clock	Beat	Script + action
0:30– 1:15	Live sense – the mesh [v9]	Dashboard shows five feed chips, four green. "Five camera feeds — four of them live phones on this table — multiplexed through one Hexagon NPU session on this laptop: fifty-plus inferences a second, INT8, and there's the badge on every message. Zone B is <i>you</i> — 0.4 people per square metre, comfortably GREEN. Not one frame leaves this room."
1:15– 2:30	KILL- SHOT	Driver starts the surge clip on Feed A. Density climbs; TTT countdown; AMBER... RED — the gate flips : RGB red, matrix arrow diverts left beside the GATE 3 card (+ steward chirp if the buzzer was secured: "that chirp is for the steward — at 100 dB of crowd, a buzzer is theatre; our crowd-facing actuators are visual"). [v9] Driver scrolls the event log on screen : "there's the decision itself — playbook P2, density 4.1, time-to-threshold 2:40, triggered by frame 4812 — full provenance, no black box." Stopwatch overlay: "Frame to red gate: 1.4 seconds. No human touched anything. " Override beat: B turns the Knob — or Driver clicks the per-gate override — "and a human can always overrule us, at the gate or from the console, logged."
2:30– 3:15	Officer loop	(If fall-confirm is in: a teammate drops in front of C1 → medical alert auto-fires.) OnePlus buzzes: dispatch, nearest officer — Officer 2 lives on our hotspot phone: "two officers on the map; it picked the closer one." C types: <i>"man collapsed near gate 2 barrier, crowd gathering"</i> → FunctionGemma → JSON, schema_valid: true → pin on the map. "270 million parameters, on the phone, offline — cell networks die at exactly these events. A misparse is a no-op, never a wrong action."
3:15– 4:00	Venue tier + resilience	Venue view: one real cluster + two SIM-labeled zones. Cloud AI 100 returns the advisory — read the English line, show Hindi + Kannada. Narrator holds up Phone-H and toggles mobile data off : "VENUE TIER OFFLINE — zones autonomous." Surge re-triggers; the gate still flips; four camera chips still green — the mesh never needed the internet. "The crowd doesn't wait for the network. Neither do we."

Clock	Beat	Script + action
4:00–4:40	Numbers	One slide: 5-feed sustained chart · NPU-vs-CPU bars · e2e p95 · TTFT/tok-s (+ the v81 NPU probe result, whichever way it went — honesty is the flex) · battery-delta per profile · "installs in three commands with zero hardware — the sim reproduces all five feeds. In production those four phones are your existing CCTV: same RTSP contract, the pipeline can't tell the difference. Per-gate BOM under ₹5,000."
4:40–5:00	Close	"We cut face recognition, the VLM, audio sensing, GPS tracking — everything that didn't reduce response latency or improve prediction. What's left is small, honest, and it works. CrowdVision: see it coming, stop it early."

Hardware-failure backup: the 2-minute recording of the full live loop on the Surface desktop *and* a personal phone; any beat that dies → play its segment, resume live at the next beat; 3:30 cue card = drop-dead jump to Numbers. [v9]
Feed failure mid-demo is pre-scripted: if a camera chip goes amber/red on stage, point at it — "and that's the watchdog being honest about a degraded feed; the zone holds safe state rather than guessing" — a failure converted to a feature, rehearsed.

Pre-rehearsed Q&A: unchanged three (cloud-only? · occlusion at crush density? · operator acceptance?) plus the energy question — answers as v8, with one v9 addition to the occlusion answer: "...and Gate 3 has a dedicated lane camera precisely because gate throughput is where counting accuracy matters most." Likely fifth [v9]: *"Are those phone cameras realistic?"* — "They're stand-ins for venue CCTV over the identical RTSP contract — in deployment you point us at the existing cameras. What the phones prove is the harder claim: one Snapdragon laptop sustains a five-camera zone on its NPU alone."

§m — Risk Register (top 8 + one live upside) [v9]

#	Risk	LxI	Mitigation	Trigger → fallback
1	QNN EP won't attach on the provided PC	MxH	Pinned wheelhouse; <code>verify_npu.py</code> with remediation text; tested	G1 red → backup export; 19:00 → CPU EP with honest badge

#	Risk	LxI	Mitigation	Trigger → fallback
			13:00 = 4 h mentor runway	
2	[v9] RTSP instability / phone thermal throttling over sustained streaming	MxM	480p@12 cap · phones on charge, screens off · per-feed watchdog in <code>capture.py</code> (backoff reconnect, stale detector, honest OK/DEGRADED/LOST) · 2 h overnight soak doubles as endurance test · charging audits at 17:00/23:00/02:30/08:30 · rehearsal feed-mix lock: minimum live = Feed A + C1 + C4; C2/C3 live only after two clean rehearsals	Feed drops on stage → the pre-scripted watchdog line (§!); zone flips UNKNOWN, gates hold state – honesty as theatre
3	Hotspot LAN instability / indoor cellular dead spot	MxH	Phone-H tested at check-in; C3 pre-cloned to identical SSID (mesh degrades to 3 cams by pre-agreement); LAN carries only ~5 Mbps	Cellular dead → cloud beats degrade to <code>template_fallback</code> (a scripted feature); LAN dead → swap to C3-hotspot, 60 s
4	Modulinos gone before we reach the table (20 each, ~50 teams)	MxM	Runner sprints distribution at 12:00; zero Modulino dependencies (flags, auto-detect)	No Knob → dashboard override beat; no Thermo → config default; no Buzzer → line dropped

#	Risk	LxI	Mitigation	Trigger → fallback
5	UNO Q Bridge API mismatch / under-power via Surface USB-C	MxM	[v9] API names pinned July 9 from User Manual + built-in examples (guide warns they vary by App Lab version); Blink + RPC echo is the <i>first</i> 13:00 action; power fallbacks: second Surface port → personal laptop port → its PD charger	Bridge red at 14:00 → mentor immediately (this is the one true unknown); gate node runs sim-rendered until fixed
6	Cloud AI 100 endpoint down/slow in the demo slot	MxM	Async, off the safety path; endpoint/model/key confirmed at check-in; template_fallback badged	Beat 4 becomes "it's offline right now — watch the zones not care"
7	FunctionGemma misparses live	MxL	Schema validation → no-op; two rehearsed utterances + one free-form; form on the same screen	"That's a no-op, never a wrong action — watch," retry once, else form
8	Demo overrun / mid-demo death / no projector · Compliance (Phase-1 deviation x2, loaner)	MxH	Two timed rehearsals · 3:30 cue card · backup video x2 locations · dashboard-is-a-URL · written scope confirmation at 10:00 covering gate lines AND the camera mesh (§7.b.ii) · loaner checklist · device return 16:00	Beat dies → play its segment; consent flags a delta → align framing same-day
†	[v9] Live upside — Sarvam session (Sat 11:30): D attends as dedicated listener for API	—	If offered: upgrades template_fallback.py → trilingual-on-edge (sarvam-edge badge) — the uplink-cut beat gets <i>stronger</i> (three languages survive the cut)	Adopt only if ≤90 min integration, decided at G2; core is never traded for it

#	Risk	LxI	Mitigation	Trigger → fallback
	access or an on-device Indic model			

Standing rules: two-strike bug rule · G4 freeze at 03:00 · E owns the clock · the MVP core is never traded for a should-have.

§n – Compliance Checklist (final pass before G6) [v9]

- [] Repo public; **MIT LICENSE** at root (Rules §8c).
- [] README: description · **all FIVE team members' names + emails – explicitly verify the fifth entry is present** (Rules §7c.ii) · setup-from-scratch incl. dependencies · run/usage.
- [] Edge-majority statement ("where the intelligence runs" table) – majority on-device, cloud additive (Rules §7c.iv).
- [] No closed-source **in the pipeline**: AOSP LocationManager, Leaflet + local floorplan, open models, MIT/Apache/EPL deps (Rules §7c.i). **[v9] Camera-app stance documented in README Notes: the RTSP streamer is an input device, not included code; pipeline consumes standard RTSP; no specific app is required.**
- [] Runnable from instructions on first attempt: fresh-clone test passed on a non-author machine (Rules §7c.iii).
- [] Store-readiness + privacy note: counts, never identities; no faces stored; streams never leave the local network (Rules §7c.v–vi).
- [] **[v9] Rules §17(c) – confidentiality: NO social-media posts about hackathon status, shortlisting, progress, or results by ANY team member without the Sponsor's prior written consent. Briefed to all five members July 10; re-briefed at check-in. This includes LinkedIn "we got shortlisted!" posts – the rule names disqualification as a consequence.**
- [] **[v9] Written scope confirmation obtained at 10:00 check-in covering both Phase-1 deltas: gate-line counting AND the personal-phone camera mesh** (Rules §7.b.ii); email trail archived.
- [] No demeaning role labels anywhere in repo, docs, commit messages, or comments (G5 audit item).
- [] Optional-but-recommended included: tests (`sim/tests`), Notes, References, commented code.

- [] Submitted via Microsoft Form **by 12:15 Sun**, confirmation screenshotted; demo A/V confirmed; devices staged for 16:00 return.
- [] Hardware doctrine audit: nothing in demo or README depends on anything beyond provided devices + personal phones/laptops + their own cables/chargers + one folded paper card.

Final cross-check (v9) [v9]

- "2 feeds / Feed 1 / Feed 2" eliminated everywhere; the mesh is Feed A (clip) + C1–C4, five feeds total, consistently in §b, §c, §d, §f, §g, §h, §i, §j, §k, §l, §m.
- §Step-3 reflects the phone-NPU probe (either outcome scored as rigor) and the honest 7-orchestrated/9-physical device count.
- §m carries the new RTSP/thermal risk with the 480p cap, charge-only streaming, watchdog, and rehearsal feed-mix lock; Sarvam is logged as a timeboxed upside.
- §0.1 opens with the complete v9 change list, including the three factual corrections (IP Webcam licensing, OnePlus arrival date vs July-10 testing, C4 feeding the PC pipeline rather than the UNO Q).

End of v9. Protect the core: sense → predict → act → inform, under two seconds, on the edge — now with five eyes on one NPU.

So — Rules §7–§8 Audit Addendum [v9.1]

Clause-by-clause re-read of Official Rules §7 and §8 against this plan. Items below were NOT explicitly addressed before this addendum; they are now binding checklist items.

A. Repository license hygiene (Rules §8.c, §8.d, §7.c.i) — the material finding.

§8.c licenses *our* Entrant Materials under MIT; it cannot and does not re-license third-party components. §8.d restricts QTI Products to the hackathon Purpose and reserves all rights. Consequences:

- [] **Never commit Qualcomm runtime binaries** to the repo: `QnnHtp.dll`, `libQnnHtp*.so`, `libLiteRtDispatch_Qualcomm.so`, `libQnnSystem.so`, etc. On the PC these arrive via the `onnxruntime-qnn` pip dependency (fine). On Android, the README instructs copying them from the official LiteRT-LM sample app per the Developer Guide — we do not redistribute them. Add these patterns to `.gitignore` on July 9.

- [] **Never commit model weights.** Gemma/FunctionGemma are under the Gemma Terms of Use (not MIT-relicensable); **YOLOv8/YOLO11 (Ultralytics) are AGPL-3.0** — open-source (so §7.c.i is satisfied) but viral, so weights and any Ultralytics code stay out of the MIT repo. `download_models.py` fetches everything at setup with license notices printed.
- [] Add **THIRD_PARTY_LICENSES.md**: Ultralytics AGPL-3.0 · Gemma Terms · Qualcomm SDK/QNN terms (via pip) · Leaflet BSD · Paho EPL · etc. Our code: MIT. This single file converts a potential judge objection into a documentation point.

B. Brought devices should be Snapdragon where possible (Rules §7.b.vi + Info Guide "bring your own Snapdragon-powered devices").

- [] Prefer Snapdragon-powered personal phones for C1–C4 when choosing which teammate's phone plays which role; regardless, the README and demo state that camera phones perform **zero compute** — they are pure RTSP sources, stand-ins for CCTV — so all intelligence demonstrably runs on Snapdragon silicon either way.

C. Venue-closing contingency (Rules §7.a vs Info Guide overnight allowance).

The Rules contemplate mandatory departure at closing time with Platforms taken to accommodation and returned next day; the Info Guide permits Saturday overnight. These conflict.

- [] Confirm at orientation which applies. **Branch if the venue closes:** loaner permits taking devices — the 03:00–06:00 soak test and sleep pods relocate to accommodation; camera phones keep charging there; return by the announced time with all Platforms. Nothing in the plan's critical path requires the venue building overnight except mains power, which accommodation also has.

D. Tie-breaking order (Rules §7.f) — strategic note.

Ties are broken by the highest score in the **first applicable criterion — Technical Implementation (40)** — then Innovation, then Deployment, then Presentation. Technical is therefore both the largest criterion *and* the tiebreaker: benchmark quality and NPU verification are the plan's insurance policy twice over. G5 triage stays README-first (protects 20 pts outright), but benchmarks are never sacrificed below "complete and committed."

E. "Must function as depicted in the text description" (Rules §7.c.v).

- [] Final README pass at G5: every claim describes what **actually runs at submission time**. If the feed-mix lock leaves C2/C3 as looped files, the

README says "2 live RTSP cameras + 3 file feeds demonstrated; pipeline supports 4+ live RTSP sources" — never the reverse. Over-claiming here is a disqualification-grade risk, not a style issue.

F. Sponsor marks (Rules §8.f).

- ☐ **No Qualcomm/Snapdragon/Hexagon logos or brand assets** anywhere in the repo, dashboard, slides, or demo video. Plain-text platform names used solely for technical description (which the submission requirements themselves demand) are fine; logos, wordmark graphics, or promotional framing are not. Pairs with the §17(c) social-media silence already in §n.

G. Minor awareness items (Rules §7.b.iv–v, §7.g, §8.c).

- ☐ Taxes on any hardware benefit or Award are the team's responsibility; be ready to furnish tax/verification documents if requested (have PAN details accessible).
- ☐ "No disruption to shared resources" — our private hotspot keeps us off shared venue internet entirely; worth saying to organizers at check-in as a courtesy.
- ☐ Be prepared to execute further license documentation if the Sponsor requests it (§8.c).

End of addendum. §n and this §o together are the pre-G6 compliance gate.